FOR A FINITE PSEUDO-COHERENT MORPHISM THE RIGHT ADJOINT OF PUSHFORWARD IS THE INTERNAL HOM OUT OF THE PUSHED FORWARD STRUCTURE SHEAF

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0AX2 (sources/stacks-duality/pinned-original-source.tex line 2763).

Allowed standard prerequisites (editorial).

- Pseudo-coherence of a finite morphism and of its pushforward of the structure sheaf. More on Morphisms, Lemmas labelled more-morphisms-lemma-Noetherian-pseudo-coherent and more-morphisms-lemma-finite-pseudo-coherent, cited at proof lines 2801-2809. Makes the internal Hom land in the required category.
- Quasi-coherence of the internal Hom of a pseudo-coherent module. Derived Categories of Schemes, Lemma labelled perfect-lemma-quasi-coherence-internal-hom, cited at proof lines 2809-2813. Gives the mapping property.
- The two adjunctions identifying the functors. Same chapter, Lemmas labelled lemma-twisted-inverse-image-bounded-below and lemma-sheafhom-affine-adjoint, cited at proof lines 2815-2819, and Derived Categories of Schemes, Lemma labelled perfect-lemma-affine-morphism-equivalence, cited in the statement. Shows both functors are right adjoint to the same functor.

Source reference. Original lemma 0A9E: sources/stacks-duality/pinned-original-source.tex, source lines 653–653. Same-chapter pointer into the author environment lemma at source line 653; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0AVW: sources/stacks-chow/tags.txt, source lines 13699–13699. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line None of the counted proof.

Date: stacks-project:duality.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

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Source reference. Original lemma 0684: sources/stacks-chow/tags.txt, source lines 7993–7993. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 2784 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0AVX: sources/stacks-chow/tags.txt, source lines 13700–13700. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 2787 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0A6H: sources/stacks-chow/tags.txt, source lines 12845–12845. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 2792 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0A9I: sources/stacks-duality/pinned-original-source.tex, source lines 726–726. Same-chapter pointer into the author environment lemma at source line 726; invoked at proof line 2797 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0BV0: sources/stacks-duality/pinned-original-source.tex, source lines 2737–2737. Same-chapter pointer into the author environment lemma at source line 2737; invoked at proof line 2798 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let $f : Y \to X$ be a finite pseudo-coherent morphism of schemes (a finite morphism of Noetherian schemes is pseudo-coherent). The functor $R\mathcal{H}om(f_*\mathcal{O}_Y, -)$ maps $D^+_{QCoh}(\mathcal{O}_X)$ into $D^+_{QCoh}(f_*\mathcal{O}_Y)$. If X is quasi-compact and quasi-separated, then the diagram*

$$\begin{array}{ccc} D^+_{QCoh}(\mathcal{O}_X) & \xrightarrow{\quad a \quad} & D^+_{QCoh}(\mathcal{O}_Y) \\ & {\scriptstyle R\mathcal{H}om(f_*\mathcal{O}_Y, -)} \searrow \quad \swarrow {\scriptstyle \Phi} & \\ & D^+_{QCoh}(f_*\mathcal{O}_Y) & \end{array}$$

is commutative, where a is the right adjoint of Lemma 0A9E for f and Φ is the equivalence of Derived Categories of Schemes, Lemma 0AVW.

Proof. (The parenthetical remark follows from More on Morphisms, Lemma 0684.) Since f is pseudo-coherent, the $\mathcal{O}_X$-module $f_*\mathcal{O}_Y$ is pseudo-coherent, see More on Morphisms, Lemma 0AVX. Thus $R\mathcal{H}om(f_*\mathcal{O}_Y, -)$ maps $D^+_{QCoh}(\mathcal{O}_X)$ into $D^+_{QCoh}(f_*\mathcal{O}_Y)$, see Derived Categories of Schemes, Lemma 0A6H. Then $\Phi \circ a$ and $R\mathcal{H}om(f_*\mathcal{O}_Y, -)$ agree on $D^+_{QCoh}(\mathcal{O}_X)$ because these functors are both right adjoint to the restriction functor $D^+_{QCoh}(f_*\mathcal{O}_Y) \to D^+_{QCoh}(\mathcal{O}_X)$. To see this use Lemmas 0A9I and 0BV0. □